GOVERNMENT OF INDIA

Ministry of New & Renewable Energy

Powering India's Clean Energy Future

Confidential and Proprietory

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SHRI NARENDRA MODI

HONORABLE PRIME MINISTER, INDIA



MESSAGE

SHRI PRALHAD VENKATESH JOSHI

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MESSAGE

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MESSAGE

The transition towards clean and sustainable energy represents one of the most significant opportunities for India's economic, environmental, and technological future. The Ministry of New and Renewable Energy has been at the forefront of driving this transformation by promoting renewable energy deployment, green technologies, energy innovation, domestic manufacturing, and investments that support India's long-term energy security and climate commitments.

The Chamber of Commerce and Industries of India (CCI India) recognizes renewable energy as a catalyst for sustainable industrial development, employment generation, technological innovation, and global competitiveness. Through policy advocacy, investment facilitation, industry collaboration, startup promotion, technology partnerships, research support, and skill development initiatives, CCI India remains committed to supporting the Ministry's vision of building a clean energy economy.

This Knowledge Paper provides comprehensive insights into the Ministry's strategic priorities, policy landscape, flagship programmes, investment opportunities, and emerging technologies across solar energy, wind power, green hydrogen, bioenergy, energy storage, smart grids, electric mobility integration, and decentralized renewable energy systems. It also identifies collaborative pathways that can accelerate innovation and private sector participation.

The publication serves as a valuable resource for policymakers, energy developers, investors, technology providers, researchers, academic institutions, entrepreneurs, financial organizations, and development partners interested in advancing India's renewable energy ecosystem. Through collaborative action and knowledge exchange, it aims to promote sustainable investments, technological excellence, and long-term energy resilience.

As India advances towards achieving its clean energy ambitions and climate goals, strong partnerships between government, industry, and innovators will remain indispensable. CCI India remains dedicated to supporting the Ministry of New and Renewable Energy in building a sustainable, resilient, and globally competitive energy future.

Munindra Kumar
Director General
CCI India

PREFACE

India's vision of becoming a **Viksit Bharat by 2047** is inseparable from its transition towards a clean, secure, and sustainable energy future. As one of the world's fastest-growing economies, India's rising energy demand must be met through reliable, affordable, and environmentally responsible sources. The **Ministry of New and Renewable Energy (MNRE)** stands at the forefront of this transformation, driving the nation's renewable energy expansion, strengthening energy security, promoting green technologies, and positioning India as a global leader in the clean energy transition.

This Knowledge Paper has been prepared by **CCI India – the Chamber of Commerce and Industry of India** as a strategic advisory document to provide a comprehensive understanding of the Ministry's institutional framework, policy priorities, flagship initiatives, investment landscape, and future growth opportunities. Drawing upon official government publications, policy documents, Budget allocations, industry reports, and international best practices, the paper presents an evidence-based assessment of India's renewable energy ecosystem while identifying opportunities for meaningful public-private collaboration.

The Ministry's mandate today extends far beyond renewable power generation. It encompasses solar and wind energy, green hydrogen, bioenergy, energy storage, domestic manufacturing, technology innovation, climate commitments, and international cooperation. As India accelerates towards its ambitious renewable energy and net-zero targets, MNRE has emerged as a key driver of economic growth, industrial competitiveness, energy independence, and sustainable development.

CCI India envisions itself as a strategic bridge connecting government, industry, investors, startups, financial institutions, research organisations, and technology providers. Through policy advocacy, investment facilitation, entrepreneurship development, technology partnerships, skill development, and stakeholder engagement, CCI India seeks to complement the Ministry's efforts in strengthening India's renewable energy ecosystem. This Knowledge Paper reflects CCI India's commitment to fostering collaborative partnerships that accelerate innovation, attract investment, and support India's transition towards a resilient and globally competitive clean energy economy.

This document is intended for Union Ministers, Secretaries to the Government of India, NITI Aayog, senior policymakers, State Governments, industry leaders, investors, research institutions, multilateral organisations, and other stakeholders seeking to engage with India's rapidly evolving renewable energy sector. It aims to serve as a strategic reference for informed decision-making while fostering long-term collaboration between the Government of India and the private sector in building a sustainable, energy-secure, and climate-resilient India.

Chamber of Commerce and Industries of India (CCI INDIA)

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About CCI India

CCI India (The Chamber of Commerce and Industry of India) is a premier national business chamber dedicated to fostering economic growth, industry development, entrepreneurship, innovation, and investment promotion across India. As the voice of Indian business and industry, CCI India serves as a bridge between government, industry, startups, academia, investors, and civil society, facilitating meaningful collaboration for nation-building and sustainable development.

Through policy advocacy, business facilitation, research, stakeholder engagement, capacity building, and strategic partnerships, CCI India works to create an enabling environment for enterprises of all sizes. The Chamber actively supports key national initiatives such as Startup India, Make in India, Digital India, Atmanirbhar Bharat, Skill India, and Viksit Bharat 2047 by connecting innovative enterprises with government opportunities and promoting inclusive economic development.

CCI India operates through sector-specific councils, industry forums, knowledge platforms, and state-level engagements that address emerging opportunities and challenges across sectors including manufacturing, agriculture, technology, healthcare, education, infrastructure, sustainability, MSMEs, and startups. The Chamber regularly organizes policy dialogues, investment summits, business delegations, workshops, conferences, and knowledge-sharing initiatives to strengthen India's economic ecosystem.

With a strong commitment to innovation, entrepreneurship, and public-private collaboration, CCI India strives to empower enterprises, facilitate investment, enhance competitiveness, and contribute to India's journey toward becoming a globally competitive and economically resilient nation. Its role as a knowledge partner and industry catalyst enables ministries, departments, state governments, startups, and businesses to collaborate effectively in achieving national development objectives.

1. Introduction to the Ministry of New and Renewable Energy

1.1 Constitutional and Legislative Mandate

The Ministry of New and Renewable Energy operates under the constitutional framework provided by Entries 53 and 54 of List I (Union List) of the Seventh Schedule, which confer upon the Union Government the authority over regulation and development of oilfields, mineral oil resources, and petroleum and petroleum products with electricity and energy policy flowing through Entry 38. The Ministry was formally constituted as a standalone entity in 2006, carved out of the erstwhile Ministry of Non-Conventional Energy Sources (MNES) which itself was established in 1992 reflecting the growing strategic weight of renewable energy in national policymaking.

The legislative scaffolding rests primarily on the Electricity Act, 2003, which created the regulatory architecture for power sector liberalisation and mandated Renewable Purchase Obligations (RPOs) on distribution companies. The Energy Conservation Act, 2001, and its 2022 Amendment further strengthened the mandate by introducing carbon credit markets, energy efficiency norms for industry, and green hydrogen obligations all of which fall within MNRE's policy domain. The National Action Plan on Climate Change (NAPCC), launched in 2008, institutionalised the National Solar Mission as one of eight missions, cementing MNRE's role as the nodal ministry for India's climate commitments.

1.2 Governance Role and Institutional Significance

MNRE is the apex policy authority for all forms of new and renewable energy in India. Its governance role encompasses policy formulation, scheme design and implementation, regulatory coordination, investment facilitation, technology development, international climate diplomacy, and capacity target setting. The Ministry works in active coordination with the Ministry of Power, Ministry of Environment, Forest and Climate Change, Ministry of Finance, NITI Aayog, and state governments, making it one of the most cross-cutting ministries in the Union Government's architecture.

The Ministry's significance transcends the energy sector. Renewable energy penetration directly affects industrial competitiveness (through lower power costs), agricultural productivity (through solar irrigation and rural electrification), urban sustainability (through building-integrated photovoltaics and smart grids), and macroeconomic stability (through reduced fossil fuel import dependence). India's crude oil import bill in FY 2024-25 exceeded USD 130 billion representing one of the single largest drains on foreign exchange reserves. Every percentage point of renewable energy substitution reduces this pressure materially.

1.3 National Significance

MNRE’s work focuses on energy security, climate justice, and economic development, aligning with Viksit Bharat 2047. India aims to be a USD 30 trillion economy by 2047, necessitating clean energy infrastructure for global competitiveness. The IEA predicts India will lead global energy demand growth by 2050. MNRE’s management of this demand through renewable capacity, technology indigenisation, and green exports will influence India’s position in the clean energy geopolitical landscape.

2. Historical Timeline of the Ministry

The institutional evolution of MNRE reflects India's shifting energy priorities across five decades. What began as a marginal policy experiment in the 1970s has become one of the most consequential ministries in the Government of India, managing a sectoral portfolio that attracts over USD 15 billion in annual investment.

Table 2.1

Period	Key Development	Strategic Significance
1981	Commission for Additional Sources of Energy (CASE) established under Department of Science & Technology	First institutional recognition of non-conventional energy as a policy priority
1982	Department of Non-Conventional Energy Sources (DNES) created	Standalone departmental identity; early focus on biogas, solar cookers, windmills
1992	MNES elevated to full Ministry status	Political recognition of renewable energy as a sovereign concern; expanded budget and mandate
2001	Energy Conservation Act enacted; Bureau of Energy Efficiency (BEE) formed under Ministry of Power	Demand-side management becomes a policy tool; early synergies with renewables agenda
2006	MNES renamed as Ministry of New and Renewable Energy (MNRE)	Institutional rebrand reflecting transition from 'non-conventional' to 'new and renewable' energy paradigm
2008	Jawaharlal Nehru National Solar Mission (JNNSM) announced under NAPCC	India's first large-scale solar commitment; 20 GW target by 2022 set the trajectory
2010	JNNSM formally launched; Feed-in Tariff (FiT) regime established	Commercial-scale solar deployment begins; viability gap funding introduced
2013	Wind energy capacity crosses 20 GW; India becomes 5th largest wind market	Wind energy matures as a utility-scale technology; private investment accelerates
2015	Solar capacity target revised from 20 GW to 100 GW by 2022; MNRE budget doubled	Paris Agreement commitment; India's climate ambition goes from incremental to transformational
2018	Reverse auction mechanism institutionalised; solar tariffs hit record low of INR 2.44/unit	Market-discovered pricing replaces administrative tariffs; private sector confidence surges
2020	National Hydrogen Mission announced; PM-KUSUM scheme scaled up	India positions for green hydrogen economy; agri-solar convergence operationalised
2021	COP26: India pledges 500 GW non-fossil capacity and net-zero by 2070	MNRE elevated to front-line of India's global diplomacy; targets become treaty-level commitments
2023	National Green Hydrogen Mission approved; PLI scheme for solar PV modules scaled	Hydrogen economy enters execution phase; domestic manufacturing ecosystem prioritised

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